Commodity Hedging & Structuring Workbench
Phase 6 — Validation & Decision Report

learner workbench (extension of the Citi MQA Forage simulation)

September 2026

Validation verdict: six of seven outcome checks PASS. The one FLAG (historical 95% VaR coverage of 38 breaches versus 25.2 expected under the static historical estimator) is retained and discussed; the filtered (EWMA-scaled) estimator remediated it, covering at 5.5% of days versus the 5% expectation (ratio 1.10, inside the pre-declared band).

This report closes the validation phase of a practitioner-style commodity hedging workbench built as an extension of the Citi MQA Forage simulation. The workbench prices and hedges a roaster's coffee purchase program on a frozen ICE Coffee C (KC=F) curve: data gates, implied convenience-yield term structure, a Schwartz–Smith two-factor fit, GARCH(1,1) working volatility, a roaster futures-and-collar hedge program, a capital-protected participation note, and a junior xVA layer (EE/EPE/PFE, unilateral CVA, netting and collateral).

Phase 6 adds two things. First, an SR 26-2-aligned validation harness: a model inventory, a fail-closed no-look-ahead gate on calibration windows, and a rolling-window outcomes program across every model. Second, the decision analysis: the roaster's outcome unhedged versus futures-only versus a zero-cost collar, with margin-funding peaks per strategy.

Headline decision numbers (1.2M lb annual purchases, 2.67 average contracts, front price 324.25 ¢/lb): the full futures hedge locks the annual purchase cost at $3,891,000 with a 95th-percentile margin peak of $103,797; the zero-cost collar keeps the cost locked between the strikes with a materially lower margin peak ($44,250 at the 95th percentile) at the price of re-exposure beyond the strikes; the unhedged roaster faces a 95th-percentile cost of $5,288,885 and a simulated worst case of $10.3M.

All regulation content (SR 26-2, IFRS 9, EMIR 3, MiFID II/PRIIPs) is *mapped as methodology*, never asserted as compliance. All numbers in this report are regenerated in one run by `report/make_numbers.py` from the frozen data (SHA-256 manifests); the hand-calculated checks of Section 3 are reproducible without reading code.

1 Scope and Honest Framing

This project extends the Citi MQA Forage simulation onto real frozen market data. It is a learner workbench, not a production system: no Citi affiliation is claimed, no internal methodology is reproduced, no regulatory compliance is asserted, and nothing here is investment advice. The data (Yahoo Finance daily bars, SOFR from FRED) is checksummed and gated; exchange-data redistribution licensing forbids committing the price bars, so only manifests, gate reports and executed notebooks are committed and the bars regenerate locally through the gated freeze. Any future cloud artifact is gated snapshots only.

2 Methodology Walkthrough (Phases 1–5)

2.1 Data gates and MQA replication (Phase 1)

The coffee chain (8 contract months plus the continuous KC=F front) is frozen with SHA-256 manifests and count/completeness gates, with a gold (GC=F) fallback armed and the pipeline fail-closed: a coffee gate failure switches universe, a gold failure stops everything. The Forage Task 3 methodology (forward curve, volatility, SOFR discounting) is re-run entirely on the frozen data; the simulation's toy numbers are labelled context only.

2.2 Carry and curve calibration (Phase 2)

As of the frozen date (2026-09-04) the curve is backwardated across every consecutive spread: front implied convenience yield 40.8%/yr decaying to 6.3%/yr on the back. The Schwartz–Smith (2000) two-factor fit in reduced form gives a mean-reversion speed $\kappa = 5.97$/yr (half-life ≈ 1.4 months), an equilibrium level $\xi = 5.673$ in logs ($e^{\xi} \approx 291$ ¢/lb), and a fit RMSE far inside the pre-declared tolerance. GARCH(1,1) working volatility: last conditional 39.2%/yr, long-run 38.5%/yr, persistence 0.802; the EWMA ($\lambda = 0.94$) reference reads 47.2%/yr. SOFR is frozen at 3.66%.

2.3 Roaster hedge program (Phase 3)

The roaster's exposure (100,000 lb/month, 12 months) maps to contract selection with the in-or-after-month rule and a 10-day pre-expiry roll schedule, averaging 2.67 contracts. A long hedge plus a zero-cost collar (long 280 put / short 382.06 call, both priced at 7.69 ¢/lb, premium gap 1.3×10^{-14} by construction) frames the cost band. IFRS 9 princip-based effectiveness (economic relationship, credit-dominance, hedge-ratio consistency) is tested, not the retired IAS 39 80–125% band; EMIR 3 NFC threshold and hedging-exemption criteria are mapped.

2.4 Structured participation note (Phase 4)

A capital-protected participation note on KC=F (notional \$1M, 9-month tenor, participation $p = 0.6$, ATM) prices in closed form as a discounted zero plus p times a Black-76 call over the reference price: $0.97292 + 0.07877 = 1.05169$ per unit notional. The antithetic MC engine (100k paths) reproduces it at 1.051913 ± 0.000452; the two-way match is the trust mechanism. Issuer-side delta/vega greeks are validated by bump-and-reprice MC.

2.5 Risk gates and junior xVA (Phase 5)

On the 2.67-contract book: daily 95% VaR \$12,900 / ES \$17,615; 99% VaR \$19,928 / ES \$25,921 (ES $\geq$ VaR asserted everywhere). The lognormal martingale exposure engine gives EPE \$33,755 and 1-year PFE \$250,975. Bucketed unilateral CVA at 3% flat hazard, 60% LGD, SOFR-flat discounting: \$583.79. Netting and collateral transforms are asserted against their invariants (netted CVA $\leq$ sum of standalones; collateral attenuates EE monotonically).

3 Hand-Calculated Verification

Per the spec, an interviewer must be able to verify the numbers without reading code. Each check below is a test in the suite with the arithmetic worked out in its docstring.

Check	Hand calculation	Result
CVA, two buckets, constant EE \$1M, hazard 10%, LGD 60%, $r = 4\%$	$dS_1 = 1 - e^{-0.05} = 0.04877058$, $df_1 = e^{-0.02} = 0.98019867$; $dS_2 = e^{-0.05} - e^{-0.10} = 0.04639219$, $df_2 = e^{-0.04} = 0.96078944$; CVA $= 0.6 \times 10^6 \times (0.04780387 + 0.04457460)$	\$55,427.08 (exact match)
VaR coverage, two 5-point windows at 80%	W1 hist $[-5..-1]$: VaR $= \lvert q_{0.2}\rvert = 4.2$, next window 0 breaches; W2 hist $[-2..2]$: VaR $= \lvert q_{0.2}\rvert = 1.2$, next window ($-10$ first) 1 breach	VaR $[4.2, 1.2]$, breaches $[0, 1]$
EWMA forecast, returns $[1, 2, 3]\%$, $\lambda = 0.94$	weights $w = (1, 0.94, 0.8836)$; weighted variance $\frac{\sum w(x-\bar{x})^2}{\sum w - \sum w^2 / \sum w} = 0.9993624$; annualised $\sqrt{252 \times 0.9993624}$	15.869%/yr
Collar net cost, synthetic zero-gap collar (f0 320, put 280, call 350), volume 100k lb	unhedged $= 1000 f_T$; futures locked at 320k; collar: 240k below the put ($f_T - K_p + f_0$), 320k between, 350k at $f_T = 380$	exact on all four scenarios
Filtered VaR (95%), 4-day price series 100/101/103/102, \$1000 per return-fraction	VaR at t uses only returns $\leq t-1$ and price($t-1$): EWMA var ($\lambda = 0.94$, weights 0.06/1) over $\{1\%, 1.9802\%\} = 0.054381/0.11321 = 0.48039\,\%^2$ $\to$ sd $0.6931\,\%$/day; VaR $= 1.6449 \times 0.006931 \times 103 \times 1000$	1,174.26 (exact match)
Collar margin cap, contracts = 2 (\$750/cent)	futures drawdowns 15k/90k; collar funding loss capped at $(320 - 280) \times 750 = 30$k	$[15k, 30k]$ capped

Table 1: Hand-calculated verification table (all reproduced exactly by the test suite).

4 SR 26-2-Aligned Validation

The posture is methodological fidelity, not compliance: the harness reproduces the *mechanics* SR 26-2 (2026-04-17, superseding SR 11-7) expects of model-risk discipline (inventory, conceptual soundness, outcomes analysis, benchmarking) over a frozen dataset, with every finding reproducible from the package.

4.1 Model inventory

4.2 Conceptual soundness

Each model's assumptions are stated in its module docstring and reproduced in the inventory above. Two deliberate simplifications are flagged rather than hidden: the Schwartz–Smith fit is reduced-form on a single snapshot (volatility parameters would need the full Kalman MLE), and the collar's margin treatment assumes the long put's gain is monetisable day-by-day (real clearing charges full futures VM daily and settles options separately).

4.3 No-look-ahead gate

Calibration windows are gated fail-closed: any observation after the valuation as-of date raises a `LookaheadError`. The historical as-of reconstructions in the outcomes program pre-truncate each frozen series to the as-of window and then gate; an untruncated (contaminated) input is refused. The gate is itself tested against a poisoned window.

Model	Inputs	Key assumptions	Benchmark	Outcomes check	Review
Data gates	frozen chain + manifests	count/completeness ceilings	SHA-256 checksums	fail-closed on injected gap	2026-09, green
Implied carry	frozen chain, SOFR	expiry ≈ 15th; storage $d = 0$	cost-of-carry identity	curve stability at past as-ofs	2026-09, green
Schwartz–Smith	curve snapshot	reduced form; κ weakly identified	NLS RMSE, $\kappa > 0$	RMSE vs tolerance	2026-09, green
GARCH(1,1) + EWMA	daily returns	constant mean	EWMA ($\lambda = 0.94$)	rolling forecast vs realized	2026-09, green
Black-76 + collar	F, K, vol, SOFR	European; discounted premiums	put-call parity; zero-cost gap	gap ≈ 0 by construction	2026-09, green
Participation note	F0, vol, SOFR, p	risk-neutral martingale futures	MC vs closed form	MC re-check	2026-09, green
IFRS 9 effectiveness	hedge P&L pairs, ratios	principles-based (no 80–125% band)	credit-dominance rejection	boundary tests	2026-09, green
Discrete-CVaR ratio	scenario P&L grid	discrete scenarios	MDPI (2025) hedge-ratio study	directional invariants	2026-09, green
GARCH-BEKK cross hedge	coffee/gold returns	diagonal BEKK(1,1)	literature hedge ratios	ratio stability	2026-09, green
Margin stress	frozen path, vol, program	monetisable-put funding cap	FSB 2024 framing; relief identities	peaks vs buffer	2026-09, green
Historical VaR/ES	realized P&L	empirical tail	ES ≥ VaR invariant	rolling breach count	2026-09, green
EE/EPE/PFE	F0, vol, SOFR, book	lognormal martingale	analytic forward MtM	invariants in suite	2026-09, green
Unilateral CVA	EE profile, hazard, SOFR	flat hazard/SOFR; no DVA	QuantLib re-derivation	CVA-vs-QuantLib re-quote	2026-09, green

Table 2: Model inventory (thirteen models, Phases 1–5; all reviewed 2026-09 with the suite green).

4.4 Outcomes analysis

Check	Finding	Status
Working vol forecast	GARCH RMSE 4.01 vs EWMA 6.80 %/yr over 4 rolling windows (ratio $0.59 \leq 1.10$)	PASS
No-look-ahead gate	active on 8 frozen series (as-of 2026-09-04); rejects a poisoned as-of window	PASS
Curve stability	front implied yield 14.8–40.8 %/yr over 6 historical as-of dates; backwardated at 4 of 6 as-ofs (two historical contango episodes)	PASS
VaR coverage (95%), static historical	38 breaches vs 25.2 expected over 4 windows (ratio 1.51, band 0.5–1.5)	FLAG
VaR coverage (95%), filtered EWMA	37 breaches vs 33.7 expected over 673 days (ratio 1.10), remediating the static FLAG	PASS
Note MC vs closed form	$\lvert\text{diff}\rvert = 8.2 \times 10^{-5}$ per unit notional, s.e. 6.2×10^{-4} (≤ 2 s.e.)	PASS
CVA vs QuantLib	relative difference 1.9×10^{-16} (same discretization, tol 10^{-9})	PASS

Table 3: Outcome-analysis findings on the frozen data.

The VaR coverage FLAG, and its remediation. The static historical 95% VaR undercovers by roughly 50% on this book: 38 breaches against 25.2 expected over four 126-day windows. Two causes were identified. First, the classic unconditional-historical-VaR failure mode on fat-tailed, volatility-clustered returns: the trailing window's empirical quantile understates the next window's tail. Second, a flaw in the backtest itself: holding one stale VaR constant across a 126-day test window conflates volatility-regime drift with model failure. The remediation is implemented in the package (ticket 13-05, `risk.filtered_var`): a filtered estimator pricing each day with the lagged EWMA variance ($\lambda = 0.94$) scaled by the lagged price. Re-run on the same book it covers at 5.5% of days versus the 5% expectation (37/673, ratio 1.10, inside the pre-declared band). The static historical estimator is retained in the findings as the baseline and its FLAG stands as the recorded evidence for why the filtered form is the production estimator of this workbench; the acceptance band was not moved.

5 Decision: Hedged vs Unhedged Outcome

Scenario engine: 50,000 lognormal martingale paths over one quarter (0.25y), the same distribution Black-76 prices. Buyer's economics: the roaster pays volume $\times f_T$ regardless; each hedge's terminal P&L reduces it. Note that the collar's strikes bound the *hedge P&L*, not the cost: between the strikes the cost is locked at the forward; below the put (280) and above the call (382.06) the net cost moves 1:1 with the market again: a crash below the put keeps helping the buyer, a spike above the call keeps hurting.

Strategy	Cost mean	Cost p95	Cost worst	Margin peak mean	Margin peak p95
Unhedged	3,895,696	5,288,885	10,296,484	—	—
Futures	**3,891,000**	3,891,000	3,891,000	43,655	103,797
Collar	3,893,639	4,595,138	9,602,737	30,647	44,250

Table 4: Annual purchase cost (1.2M lb) and margin-funding peaks (USD), 2.67 average contracts, put 280 / call 382.06, 0.25y horizon.

Reading the table. The full futures hedge is a strict price lock at \$3,891,000 (= 1.2M $\times$ 3.2425); its residual risk is purely margin funding: mean peak \$43.7k, 95th percentile \$103.8k, comfortably inside the \$800k reference-collateral assumption carried from the Starbucks 10-K framing. The collar costs almost the same on average (\$3.894M) but trades tail exposure for liquidity: its margin peak is capped at the put protection ($44{,}250 = (324.25 - 280) \times \frac{8}{3} \times 375$), roughly 58% lower at the 95th percentile, while the cost re-floats 1:1 beyond the strikes (p95 \$4.595M, worst \$9.6M in-simulation). The unhedged roaster owns the entire distribution: p95 \$5.29M, simulated worst \$10.3M.

Decision. For a roaster whose binding constraint is cash-flow certainty (the Starbucks 10-K template: \$37.9M of margin collateral posted as of FY2025), the choice is between the futures lock (price certainty, full margin exposure) and the collar (locked cost band, half the margin peak, tail re-exposure). The workbench's numbers support the collar for a margin-constrained buyer and the futures lock when budget certainty dominates, with the IFRS 9 cash-flow-hedge designation and effectiveness testing identical in structure for both.

6 Limitations

- **Basis risk is not modelled.** All hedges are evaluated on the front KC=F series; the roaster's physical purchases settle on a differential basis to the exchange. A basis-scenario overlay is the natural next step.
- **Historical volatility only.** The working vol is a GARCH(1,1) point estimate on frozen history; option premiums (collar legs, the note) inherit its parameter risk. No implied-vol surface is available for free coffee options.
- **VaR remediation is EWMA-level.** The static historical VaR's under-coverage is remediated by the filtered (EWMA $\lambda = 0.94$) estimator (Section 4); a GARCH-conditional-vol

scaling would be the next refinement but is not implemented. The static estimator is kept only as the flagged baseline.

- **Unofficial source.** The Citi MQA Forage simulation is a learning artifact, not a source of Citi methodology; all toy numbers from it are labelled context only.

- **Junior xVA only.** Unilateral CVA with flat hazard and flat SOFR; no DVA/FVA/MVA, no wrong-way-risk modelling beyond a qualitative discussion (a buyer short a call on coffee is long the reference asset; WWR here is benign by inspection).

- **Margin conventions are assumptions.** $8k initial margin per contract and the monetisable-put funding cap are stated learner assumptions, not exchange feeds.